# MAE 3127 - Fluid Mechanics Laboratory
# Recitation-0

(P-Set00: Total points = 100)

____________________________,__________________________

Last Name First Name

_________________

Date

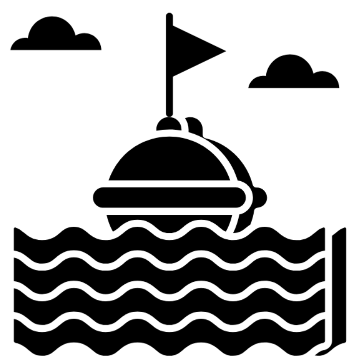


Figure 1: Buoy by Rabiya Anwer from Buoy Icons, Noun Project (CC BY 3.0)

**According to the GWU Code of Academic Integrity, I pledge that I have neither given nor received unauthorized assistance on this work.**

____________________________,_________________

Student's Signature Date

# P-Set00: Interpretation of the Tin Foil Boat experiment performed on August 27, 2025.

**R0-P000: Relationship Observed** The attached plot shows a scatter of boat volume versus number of pennies used, with a fitted trend line, suggesting a mild positive correlation between the two variables.

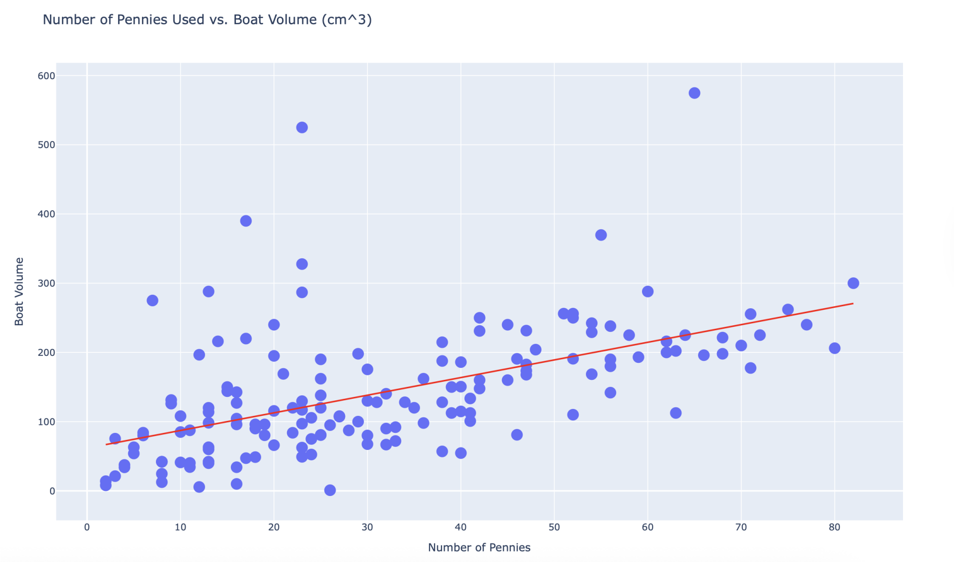


Figure 2: Tin Foil Boat Data

Larger boats should be able to support ______________________________ due to their increased buoyant force.

**R0-P001: Data Spread and Outliers** The data points are spread widely and not tightly clustered, indicating considerable variability for similar numbers of pennies and volumes. Outliers are visible, with a few boats able to support an unusually high number of pennies for their volume, or conversely, volumes much larger than others supporting similar numbers of pennies. Some boats with similar volumes can support very different numbers of pennies.

**What are the other factors that play a significant role in the variable in the data shown in Figure 2?** State at least three factors in the space provided below:

1. ____________________________________________
2. ____________________________________________
3. ____________________________________________

**R0-P002: Run the following Python program to calculate the correlation IMPORTANT:**

1. You are not required to submit this program.
2. The Python program will be discussed during the recitation
3. You will be provided instructions to install any missing libraries during the recitation or during office hours.
4. You are required to be connected to the network to install any missing libraries

Listing 1: Python code for plotting and correlation

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr

# Load data
df = pd.read_csv('boat_data.csv')

# Scatter plot
plt.figure(figsize=(8,6))
plt.scatter(df['Number of Pennies'], df['Boat Volume'], alpha=0.7)
plt.title('Number of Pennies vs Boat Volume')
plt.xlabel('Number of Pennies')
plt.ylabel('Boat Volume')
plt.grid(True)
plt.tight_layout()
plt.show()

# Calculate Pearson correlation
correlation, p_value = pearsonr(df['Number of Pennies'], df['Boat Volume'])
print('Correlation Coefficient:', correlation)
print('P-value:', p_value)
```

The positive correlation supports the general principle of buoyancy: larger volume usually allows for greater load capacity.

**The value of correlation calculated using the Python program is** __________________________.